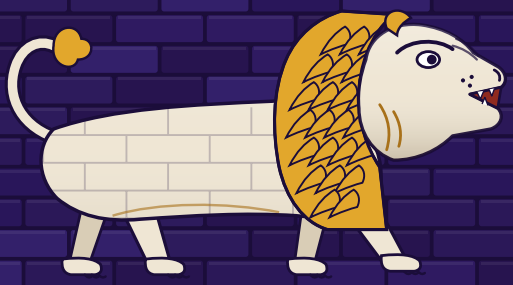
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

LS

INSTRUMENTS AND LAB SAFETY

A handout of ideas, facts and problems



8 sections · Edition of November 2026 · 27 pages

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How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Instruments and Lab Safety

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

1 Hazard, risk and prevention

Aims. Distinguish a hazard from the risk of harm in a particular situation; Choose a control that interrupts the actual exposure route.
Before this section: Hazard means potential for harm; exposure means contact with the harmful agent.

A hazard is a source of possible harm. Risk depends on exposure and the chance and severity of harm. Accident prevention is shared: prepare, follow approved procedures and report unsafe conditions. A familiar chemical or routine experiment can still present hazards.

Idea 1. A substance hazard is its potential to cause harm; the practical risk depends on how it is used and who can be exposed. Identifying the chemical is therefore only the start. A useful assessment links the harmful property to an activity, an exposure route and a suitable control.

Fact 1.

Good answer: hazard → exposure route → control

Why it holds. The same hazard can create different risks under different exposure conditions. A control is relevant only if it reduces the source, exposure likelihood or harmful consequence.

Method 1.

1. Name the harmful property and the activity creating exposure.
2. Identify who could be exposed, by which route and under what conditions.
3. Select a control that reduces the source or interrupts the route.
4. Explain the remaining need for the approved procedure and required personal protection.

Recognising it. For an unfamiliar reagent, preparation means identifying hazards and the approved method before beginning work.

Pr 1. A volatile harmful liquid on an open bench creates an inhalation hazard.

Solution. The example liquid is volatile and harmful, so vapour can leave an open container. A person nearby may inhale that vapour; this connects the chemical hazard to an exposure route. An approved fume-hood procedure controls vapour at its source when

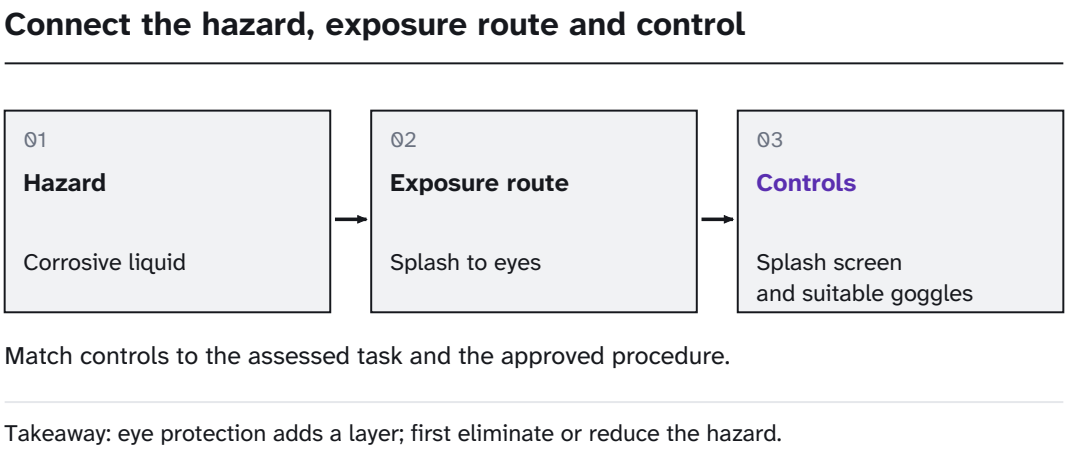


Figure 1. Figure 06. A useful safety answer connects a specific hazard to a route of exposure and suitable controls. A splash screen and appropriate goggles can limit splash exposure; they do not replace reducing or removing the hazard where possible.

suitable for that substance and operation. Goggles protect eyes from appropriate hazards but do not remove vapour from inhaled air.

Common error. Do not write only be careful; name a concrete control.

In the examination. For a scenario answer write a linked chain: hazard, exposure route, suitable control and reason.

Pr 2. Why is naming only a hazardous chemical insufficient to describe a risk scenario?

Pr 3. A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

Pr 4. The same harmful volatile liquid is sealed in approved storage or poured openly during a task. Why can risk differ?

(a) The opportunity for exposure differs

(b) Its chemical identity necessarily changes

(c) A sealed container makes the liquid chemically harmless

2 Personal protective equipment and contamination

Aims. Trace how contaminated gloves transfer material to clean surfaces; Select PPE for the task and remove it without spreading contamination.

Before this section: Contamination transfer; clean and contaminated work zones; task-specific protection.

Splash goggles protect against chemical splashes; ordinary spectacles do not provide equivalent coverage. Gloves must be compatible with the chemical and inspected. No glove protects indefinitely. Contaminated gloves can carry chemicals to phones, door handles and notebooks.

Idea 2. PPE is selected for the route and type of exposure, not simply worn as a general safety symbol. Ordinary prescription glasses leave openings around the eyes, while a coat mainly protects covered skin and clothing. Protection for a splash task must match the actual eye and body hazards specified by the method.

Fact 2.

Select PPE for the specific hazard and task

Why it holds. A barrier cannot protect an area it does not cover adequately. Different materials and tasks can require different PPE, so everyday glasses are not automatically splash-rated eye protection.

Method 2.

1. Identify the contaminant and which surfaces or body areas need protection.
2. Use PPE specified for the material and task rather than assuming one glove type protects against everything.
3. Keep contaminated gloves away from phones, door handles and other clean shared items.
4. Remove and dispose of PPE according to the approved procedure, then perform required hand hygiene.

Recognising it. When asked whether clothing is sufficient, identify the exposed area and required protection for the particular task.

Pr 5. A student touches a phone with contaminated gloves.

Solution. The student glove is already contaminated before touching the phone. Contact transfers material from glove to phone, creating a new contaminated surface. Later bare-hand contact can move that material to skin, food or the face. Removing gloves correctly before touching clean items breaks that transfer chain; the phone may still need the approved decontamination procedure.

Common error. More layers of unsuitable gloves do not establish chemical compatibility.

In the examination. Explain what is transferred and to where; simply saying wear gloves does not solve a contamination scenario.

Pr 6. Do gloves make it acceptable to touch your face during laboratory work?

Pr 7. Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

Pr 8. Why should glove material be selected for the chemical and task?

(a) Thicker-looking gloves always remove all exposure routes

(b) Different glove materials resist different chemicals differently

(c) Every glove blocks every chemical equally

3 Exposure routes and toxic effects

Aims. Identify inhalation, ingestion, skin or eye contact and injection exposures; Distinguish route of exposure from the timing of a toxic effect.

Before this section: Contact with an agent can occur by several routes; visible vapour is not required.

Chemicals can enter by inhalation, ingestion, skin or eye absorption, or injection through contaminated sharps. Acute exposure is short-term; chronic exposure is repeated or long-term. Effects can be local or systemic, immediate or delayed, reversible or irreversible.

Idea 3. Exposure route describes how material reaches the body, not whether the person notices it. Vapour can be inhaled, while a contaminated sharp can introduce material through skin. Odour is a sensory property with limits, so absence of smell cannot establish absence of a hazardous exposure.

Fact 3.

Dose, duration, route and substance all influence harm

Why it holds. Different routes require different preventive controls. Some hazards have little warning odour, and smell may not reliably indicate a safe concentration.

Method 3.

1. Read the activity and locate the point where the agent could enter or contact the body.
2. Name the route using the specific evidence in the scenario.
3. Distinguish a rapid acute effect from an effect associated with longer or repeated exposure.
4. Match prevention to the identified route and follow the material-specific procedure.

Recognising it. Classify the physical entry mechanism first, then explain why sensory impressions cannot replace safety information.

Pr 9. Eating in a laboratory can lead to ingestion of contamination.

Solution. Eating in the laboratory can move contamination from hands or surfaces to food. Swallowing that material is ingestion, even when the contaminant is not visible. A contaminated needle can breach the skin and introduce material directly: this is injection exposure. The absence of visible vapour does not remove either risk because neither route depends on inhalation.

Common error. A lack of smell does not prove safety.

In the examination. Name both the route and the evidence, for example ingestion because contaminated food is swallowed.

Pr 10. A droplet splashes into an eye. Is the route inhalation?

Pr 11. Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

Pr 12. A hazardous substance has no noticeable odour during use. What can be concluded?

(a) There is no inhalation hazard

(b) PPE and handling controls are unnecessary

(c) Odour alone cannot establish safety

4 Labels, SDS and chemical properties

Aims. Use a label and SDS to connect chemical properties with controls; Distinguish corrosive and oxidizing hazards in an explanation.
Before this section: Chemical identity; label symbols; an SDS gives substance-specific hazard and handling information.

Read the identity and hazard information before use. Safety data sheets describe hazards, handling, storage, exposure controls, first aid and other substance-specific information. Flammable, oxidizing, corrosive and toxic refer to different hazards and can apply to one substance simultaneously.

Idea 4. One chemical can have several hazards that require separate reasoning. Flammability concerns ignition and combustion, whereas corrosivity concerns damaging contact. An answer that names one generic precaution may miss the second hazard, even when it sounds broadly cautious.

Fact 4.

$$\text{Identity} + \text{hazard} + \text{controls} + \text{incompatibilities}$$

Why it holds. An ignition control addresses a fire pathway but does not by itself prevent corrosive contact. Splash/contact controls address exposed tissues but do not make a flammable material nonflammable.

Method 4.

1. Confirm the chemical identity from its label before selecting a procedure.
2. Read relevant SDS information for hazards, incompatibilities and required handling controls.
3. Explain how the named property causes harm in the scenario.
4. Use the approved storage or handling arrangement rather than inferring compatibility from appearance.

Recognising it. When a label carries more than one hazard, give a distinct risk-and-control link for each.

Pr 13. A corrosive liquid can damage tissue, while an oxidizer may intensify fire even when it is not itself a fuel.

Solution. A corrosive liquid can damage living tissue or certain materials by contact. An oxidizer can support or intensify combustion even when it is not itself the fuel. The mechanisms differ, so the same general statement handle carefully does not explain either control. A good answer links splash/contact controls to corrosivity and incompatibility or fire-prevention controls to oxidizing behaviour as specified by the SDS.

Common error. Do not infer that all colourless liquids are harmless or that one symbol captures every hazard.

In the examination. Pair each property with its consequence; listing pictogram names alone gives less evidence of understanding.

Pr 14. Does a chemical being nonflammable prove it cannot intensify a fire?

Pr 15. A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

Pr 16. A bottle is both flammable and corrosive. Which statement is best?

(a) Both fire and damaging-contact risks need suitable controls

(b) Removing ignition sources alone resolves every hazard

(c) Eye protection alone makes ignition impossible

5 Glassware, measurement and safe apparatus

Aims. Select apparatus according to measurement purpose and safe operating limits; Explain why volumetric preparation and heating require different vessels.
Before this section: Volume; calibration mark; damaged glassware can fail during use.

Inspect glassware for chips and cracks and apparatus for damaged wiring before use. A beaker is useful for holding and mixing; a graduated cylinder gives more useful volume estimates; a volumetric pipette or flask is designed for a specified accurate volume. Read liquid levels at eye height using the appropriate meniscus.

Idea 5. Accurate liquid handling depends on both choosing the right instrument and using it as calibrated. A fixed-volume volumetric pipette is intended to deliver one specified volume under its stated conditions. Safe pipetting and proper reading matter as much as the printed volume.

Fact 5.

Choose the instrument for required precision, not convenience

Why it holds. A vessel marked for approximate volume may not deliver the required fixed volume accurately. The calibration method determines how the instrument should be filled, read and drained.

Method 5.

1. State whether the task requires approximate volume, accurate delivery or preparation to a final volume.
2. Choose the apparatus designed for that purpose and check its condition.
3. Follow the specified filling, meniscus and operating procedure.
4. Before heating, verify that the vessel and setup are approved for the process.

Recognising it. For an accurate fixed transfer, distinguish a volumetric pipette from a flask that contains a final prepared volume.

Pr 17. A volumetric flask prepares a solution to its marked final volume.

Solution. The volumetric flask is calibrated to contain a stated final volume at its reference conditions. The solution is brought to the mark as part of an approved preparation procedure. Its main role is volume accuracy, not serving as a directly heated reaction vessel. For heating a reaction, use the vessel and support specified by the laboratory method rather than substituting the flask.

Common error. Extra decimal places do not make an imprecise instrument accurate.

In the examination. In apparatus questions state the purpose and limitation together: prepares final volume, unsuitable for directly heating the reaction.

Pr 18. Does having a calibration mark mean a flask is suitable for direct heating?

Pr 19. Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

Pr 20. A volumetric flask and a volumetric pipette both show 10 mL. Why are they not automatically interchangeable?

(a) The same printed number guarantees identical delivery

(b) They are designed for different volume-handling purposes

(c) Only a flask ever has calibration

6 Hoods, heating, cooling and pressure

Aims. Explain why heating a closed unsuitable vessel can create a pressure hazard; Recognize when an apparatus setup requires instructor review.

Before this section: Heating can increase vapour production and gas pressure; equipment has specific operating limits.

A fume hood controls airborne chemical exposure when used correctly. It is not a storage cabinet or a waste-evaporation device. Heating and cooling systems can introduce burns, pressure changes and material failure. Approved setups account for both temperature and pressure.

Idea 6. A heated closed vessel can store pressure as well as thermal energy. A volatile liquid supplies vapour, and heating can increase that vapour pressure. If the vessel or closure is unsuitable, failure can release hot hazardous contents suddenly; a hood does not turn it into pressure-rated equipment.

Fact 6.

Control vapour, ignition, pressure and thermal contact separately

Why it holds. Heating an enclosed volatile liquid can raise internal pressure. Container failure can eject hot liquid or vapour, adding exposure and burn consequences.

Method 6.

1. Identify heat source, volatile materials and any sealed volume.
2. Check the approved procedure and apparatus pressure and temperature limits.
3. Use the specified containment, heating and ventilation arrangements.
4. If the setup differs or becomes unsafe, stop and alert the responsible instructor instead of modifying it ad hoc.

Recognising it. A tightly sealed volatile-liquid heating setup requires checking the approved apparatus rather than improvising a modification.

Pr 21. A volatile solvent warmed in an unsuitable sealed container can build pressure.

Solution. Warming the volatile solvent increases vapour formation. In an unsuitable sealed container, expanding gas and vapour can increase internal pressure. The container or closure may fail because it was not designed for that operation. The example therefore calls for stopping and consulting the approved setup; improvising a vent is not a validated correction.

Common error. A hood does not remove every fire, pressure or contact hazard.

In the examination. Explain the mechanism, pressure buildup from heating, before giving the required stop-and-consult response.

Pr 22. Does placing an unsuitable sealed heated vessel in a fume hood solve its pressure hazard?

Pr 23. Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

Pr 24. Does placing an unsuitable sealed heated vessel in a fume hood make the pressure setup acceptable?

(a) Yes; all hoods prevent pressure buildup

(b) Yes; a hood removes liquid volatility

(c) No; ventilation does not make a vessel pressure-rated

7 Waste, housekeeping and emergency priorities

Aims. Use identification and approved procedures to choose a waste or spill response; State safe priorities when a spill material is unknown.

Before this section: Chemical incompatibility; designated waste streams; laboratory emergency instructions.

Waste must follow the laboratory segregation and disposal procedure. Broken glass, sharps and chemical waste are different streams. Good housekeeping keeps exits and emergency equipment accessible. In an incident, alert others and follow the established response; do not improvise a hazardous cleanup.

Idea 7. Emergency equipment is useful only when people can reach and operate it promptly. A storage box can create risk without being chemically hazardous itself because it delays the response to another event. Housekeeping therefore protects the reliability of the laboratory safety system.

Fact 7.

$$\text{Recognize} \rightarrow \text{alert} \rightarrow \text{follow trained response}$$

Why it holds. A blocked eyewash can delay response to an eye exposure. Keeping the access route clear reduces that delay before an emergency occurs.

Method 7.

1. Identify the material only if that can be done safely from available records or labels.
2. Keep others clear and alert the instructor or responsible responder.
3. Use the laboratory response plan for the identified hazard and your training level.
4. For routine waste, use the designated compatible container and labelling procedure.

Recognising it. An obstructed emergency station is a prevention and access problem, even when no spill or exposure is currently happening.

Pr 25. An unknown spill is not an invitation to mix in a neutralizer.

Solution. An unknown spill has uncertain properties, so an assumed neutralizer may be incompatible. Adding another chemical could cause heat, gas formation or another hazardous reaction. Alerting the instructor and keeping people clear limits exposure while the material and response are assessed. Cleanup then follows the approved spill procedure for the identified material and trained personnel.

Common error. Do not recommend that an untrained student fight every fire or clean every spill.

In the examination. In an emergency scenario prioritize alerting and exposure prevention; do not invent a chemical cleanup recipe.

Pr 26. Should an unknown laboratory liquid automatically be washed into a sink?

Pr 27. A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

Pr 28. Why can a harmless empty box near emergency equipment still create risk?

(a) It can delay access when the equipment is needed

(b) It must release toxic vapour

(c) Emergency equipment works only when boxes are present

8 Biosafety and the control hierarchy

Aims. Classify controls within the hazard-control hierarchy; Distinguish engineering containment from training, work practices and PPE.
Before this section: A biological hazard may be spread through aerosols, contact or sharps depending on the activity.

Biosafety aims to prevent unintended exposure to biological agents and their release. The general control hierarchy prefers removing a hazard, then using a safer substitute, then engineering controls, administrative controls and work practices, and PPE. The suitable controls depend on the task: removing or replacing the biological material may not be possible. Training and procedures are administrative controls; suitable containment equipment is an engineering control. Safe work practices include hand hygiene and avoiding aerosol generation.

Idea 8. Controls work at different places in the hazard pathway. Equipment containment can isolate the source, procedures guide behaviour, and PPE forms a barrier worn by a person. Classifying a control means explaining its mechanism; it does not mean one category makes every other category unnecessary.

Fact 8.

Eliminate → substitute → engineering → administrative/work practices → PPE

Why it holds. Engineering controls act through equipment or physical containment. Training and handwashing procedures guide work practices, while goggles are worn personal protection.

Method 8.

1. Identify the biological material and activity-specific exposure route.
2. Consider elimination or substitution where feasible before downstream controls.
3. Classify equipment containment as engineering, training and procedures as administrative or work-practice controls, and worn barriers as PPE.
4. Explain how the selected combination addresses the task and follows institutional biosafety procedures.

Recognising it. When asked to classify controls, describe what each does before assigning its hierarchy category.

Pr 29. Training is administrative.

Solution. Training changes how people work, so it is an administrative control. A suitable biological safety cabinet provides engineering containment for its approved uses. Gloves form a worn barrier and are PPE. A safe handling technique is a work practice, generally grouped with administrative controls; the categories describe different protective mechanisms.

Common error. A device with airflow is not automatically suitable for every chemical or biological hazard.

In the examination. Do not label every cabinet or hood as interchangeable containment; state suitable equipment for the defined biological task.

Pr 30. What is the general hierarchy from source-focused to last-line protection?

Pr 31. Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

Pr 32. A suitable containment cabinet and goggles are used together. Why can both be needed?

(a) PPE always eliminates the source hazard

(b) They interrupt exposure through different protective mechanisms

(c) They belong to exactly the same control category

Fact 9. The control hierarchy in the correct order. The hierarchy ranks ways to control a hazard. A list of control types is not automatically in the preferred order.

1. Eliminate: remove the hazard.
2. Substitute: use a safer alternative after checking its risks.
3. Engineering controls: separate people from the hazard using suitable equipment or containment.
4. Administrative controls and work practices: use training, procedures and safe methods.
5. PPE: use suitable worn protection. It adds protection but depends on correct selection and use.
6. In a laboratory, eliminating or replacing the material may not be possible. The assessed task determines the suitable combination of controls.

Pr 33. Classify a suitable containment cabinet, safety training, gloves and correct handwashing.

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Identify what the student has not established before starting: hazards and suitable controls. (2) Choose one preparatory action that obtains reagent-specific safety information. (3) Choose another that confirms the approved procedure or required supervision. Explain how each prevents an uninformed exposure.

Pr 7 (p. 5) (1) List which areas a lab coat covers and which areas remain exposed. (2) Consider whether ordinary prescription glasses provide the task-required splash protection around the eyes. (3) Explain the coverage gaps and identify the need for approved splash-rated protection and any other task-specific PPE, following the laboratory method.

Pr 11 (p. 6) (1) For each event, identify how the material crosses into or contacts the body. (2) Airborne vapour enters through breathing; a contaminated puncture breaches skin directly. Name the route for each. (3) For the smell explanation, state that detectability and hazardous exposure are different questions; refer to substance-specific safety information.

Pr 15 (p. 7) (1) Separate the two label properties before suggesting controls. (2) For flammability, explain an ignition/fire risk and a suitable approved ignition-control principle. (3) For corrosivity, explain tissue-damage risk and an approved exposure-prevention principle. Do not treat one control as covering both mechanisms.

Pr 19 (p. 9) (1) The task asks to transfer a fixed accurate volume, not prepare a solution to a final flask mark. (2) Select an appropriate fixed-volume volumetric pipette and approved pipette aid. (3) For good practices, explain correct meniscus reading and use according to its calibration/draining instructions. Mouth pipetting is not acceptable.

Pr 23 (p. 10) (1) Trace how heating affects vapour inside a tightly sealed space. (2) Connect the resulting pressure to possible vessel or closure failure. (3) State a second consequence involving hot released contents, such as burns or hazardous exposure, and relate it to the need for the approved setup.

Pr 27 (p. 11) (1) Identify the purpose of an eyewash and when rapid access matters. (2) Explain how a box could delay a person trying to respond to an eye exposure. (3) Propose promptly clearing and maintaining the route through the responsible housekeeping procedure. Link the action to access rather than claiming the box itself is corrosive.

Pr 31 (p. 12) (1) Sort the examples into equipment, behaviour/procedure and worn protection. (2) Classify the suitable cabinet as engineering containment and goggles as PPE. (3) Place training under administrative controls and correct handwashing under work practices, commonly grouped with administrative controls. Explain the basis rather than just listing labels.

Pr 33 (p. 13) (1) The cabinet is an engineering control. (2) Training is administrative; gloves are PPE. (3) Handwashing is a safe work practice, often grouped with administrative controls.

Answers

Pr 2 (p. 3) Risk also depends on how exposure can occur, how likely it is and how serious its consequences could be.

Pr 3 (p. 3) Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action.

Pr 4 (p. 3) (a) Correct. Handling conditions can change exposure while the underlying hazardous property remains.

Pr 6 (p. 5) No. Gloves can carry contamination directly to the face.

Pr 7 (p. 5) Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control.

Pr 8 (p. 5) (b) Correct. Compatibility and performance depend on material, chemical and conditions.

Pr 10 (p. 6) No. It is eye or mucous-membrane contact.

Pr 11 (p. 6) Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation.

Pr 12 (p. 6) (c) Correct. Safety depends on identified properties and exposure controls, not smell alone.

Pr 14 (p. 7) No. An oxidizer may intensify combustion without being a fuel.

Pr 15 (p. 7) Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control.

Pr 16 (p. 7) (a) Correct. The properties create separate pathways requiring an appropriate combination of controls.

Pr 18 (p. 9) No. Calibration and suitability for heating are different design requirements.

Pr 19 (p. 9) Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices.

Pr 20 (p. 9) (b) Correct. A flask contains a prepared volume; a pipette is designed to transfer a calibrated volume according to its type.

Pr 22 (p. 10) No. A hood does not make a vessel pressure-rated or validate an unsafe heating setup.

Pr 23 (p. 10) Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences.

Pr 24 (p. 10) (c) Correct. Engineering ventilation and pressure suitability are different requirements.

Pr 26 (p. 11) No. Its identity, compatibility and approved disposal route must be established.

Pr 27 (p. 11) It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention.

Pr 28 (p. 11) (a) Correct. Obstruction creates a response-time hazard even if the box material is harmless.

Pr 30 (p. 12) Elimination, substitution, engineering controls, administrative controls and work practices, then PPE.

Pr 31 (p. 12) Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each.

Pr 32 (p. 12) (b) Correct. Containment and worn barriers can form complementary protections specified for the task.

Pr 33 (p. 13) Engineering; administrative; PPE; work practice, respectively. Engineering controls come before administrative controls and PPE in the general hierarchy.

Instruments and Lab Safety, Test A

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which best describes risk?

   (a) The mass of any bottle
   (b) The same as smell
   (c) The label colour alone
   (d) The chance and severity of harm under the exposure conditions

2. Which is appropriate before touching a clean door handle?

   (a) Keep contaminated gloves on
   (b) Wipe gloves on clothing
   (c) Remove contaminated gloves correctly
   (d) Smell the gloves

3. Eating with contaminated hands is mainly which exposure route?

   (a) Inhalation
   (b) Injection
   (c) Noise exposure
   (d) Ingestion

4. An oxidizer is especially associated with:

   (a) Intensifying combustion
   (b) Preventing all fires
   (c) Measuring temperature
   (d) Making all chemicals inert

Part II. Show your reasoning.

1. A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

2. Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

3. Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

4. A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

Instruments and Lab Safety, Test B

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which is designed to prepare one accurately specified solution volume?

   (a) A beaker
   (b) A watch glass
   (c) A volumetric flask
   (d) A test-tube rack

2. A fume hood should be used as:

   (a) Permanent chemical storage
   (b) A controlled work enclosure for appropriate vapour hazards
   (c) A way to evaporate all waste
   (d) A food cupboard

3. An unknown chemical spill should first prompt a student to:

   (a) Alert the instructor and keep people clear
   (b) Mix it with another chemical
   (c) Ignore it
   (d) Taste it

4. Which is an administrative control?

   (a) Gloves
   (b) Training and written procedures
   (c) A sealed rotor
   (d) Splash goggles

Part II. Show your reasoning.

1. Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

2. Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

3. A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

4. Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

Instruments and Lab Safety, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (d) Risk depends on both the hazard and how people are exposed.

Written part. Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action. (Section: p. 2.)

Question 2. (c) This prevents spreading contamination outside the work area.

Written part. Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control. (Section: p. 4.)

Question 3. (d) Material can pass from the hands to food and the digestive system.

Written part. Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation. (Section: p. 5.)

Question 4. (a) Oxidizers can support or intensify combustion.

Written part. Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control. (Section: p. 6.)

Instruments and Lab Safety, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) The calibration mark defines the specified contained volume under its calibration conditions.

Written part. Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices. (Section: p. 8.)

Question 2. (b) It is an engineering control for relevant airborne exposures.

Written part. Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences. (Section: p. 9.)

Question 3. (a) Unknown hazards require assessment and the established response.

Written part. It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention. (Section: p. 10.)

Question 4. (b) Administrative controls organize how work is authorized and performed.

Written part. Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each. (Section: p. 11.)

Facts at a glance

1. Hazard, risk and prevention p. 2

Good answer: hazard → exposure route → control

2. Personal protective equipment and contamination p. 4

Select PPE for the specific hazard and task

3. Exposure routes and toxic effects p. 5

Dose, duration, route and substance all influence harm

4. Labels, SDS and chemical properties p. 6

Identity + hazard + controls + incompatibilities

5. Glassware, measurement and safe apparatus p. 8

Choose the instrument for required precision, not convenience

6. Hoods, heating, cooling and pressure p. 9

Control vapour, ignition, pressure and thermal contact separately

7. Waste, housekeeping and emergency priorities p. 10

Recognize → alert → follow trained response

8. Biosafety and the control hierarchy p. 11

Eliminate → substitute → engineering → administrative/work practices → PPE

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

Sources

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

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Latin letters first, then Greek letters, then differences (Δ). The same letter can name different quantities in different sections.